



JAVA ROADMAP

Step by Step to Become a Java Developer



Learn. Practice. Build. Master Java.



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Java Basics

- Setup Java Environment (JDK, IDE)
- Syntax, Variables, Data Types, Operators
- Input/Output, Type Casting
- Control Statements (if, switch, loops)

1-2 Weeks

OUTCOME

Write simple Java programs.



OOP Concepts

- Classes & Objects
- Constructors
- Inheritance
- Polymorphism, Abstraction, Encapsulation

2-3 Weeks

OUTCOME

Understand and apply Object-Oriented Programming.



Core Java – Intermediate

- Arrays & ArrayList
- String, StringBuilder, StringBuffer
- Exception Handling
- File Handling, Packages, Access Modifiers

2 Weeks

OUTCOME

Build robust programs using core Java features.



Advanced Java

- Collection Framework (List, Set, Map)
- Generics
- Lambda Expressions
- Stream API, Optional, Date & Time API

2-3 Weeks

OUTCOME

Write clean, efficient and modern Java applications.



Database & JDBC

- RDBMS Basics (MySQL / PostgreSQL)
- SQL (CRUD, Joins, Constraints)
- JDBC Basics
- PreparedStatement, Connection Pool

2 Weeks

OUTCOME

Connect Java apps with databases.



Frameworks (Start with One)

- Spring Core / Spring Boot Basics
- Dependency Injection
- RESTful API Development
- Validation, Exception Handling

3-4 Weeks

OUTCOME

Build REST APIs using a Java framework.



Testing

- Unit Testing with JUnit
- Mockito Basics
- Integration Testing

1-2 Weeks

OUTCOME

Write tested and reliable code.



Build & Tools

- Maven / Gradle Basics
- Project Structure
- Logging (Log4j / SLF4J)
- Debugging Techniques

1-2 Weeks

OUTCOME

Manage and build Java projects efficiently.



Deployment & DevOps Basics

- Build JAR / WAR
- Deploy to Server (Tomcat / Spring Boot)
- Basics of Git & GitHub
- CI/CD Basics (Jenkins / GitHub Actions)

1-2 Weeks

OUTCOME

Deploy Java applications confidently.



Real-World Projects

- Apply all concepts
- Build 2-3 real-world projects
- Document and showcase your work

Ongoing

OUTCOME

Strong portfolio that gets you hired.



ESSENTIAL TOOLS



PROJECT IDEAS

- ✓ Student Management System
- ✓ E-commerce REST API
- ✓ Blog Application
- ✓ Online Book Store
- ✓ Employee Management System

TIPS FOR SUCCESS

- ✓ Practice coding daily
- ✓ Solve DSA problems
- ✓ Build projects
- ✓ Read documentation
- ✓ Stay consistent and curious



Consistency is the key! Keep learning, keep building, and enjoy the journey.

